



CAT-Net: Convolution, attention, and transformer based network for single-lead ECG arrhythmia classification

Md Rabiul Islam ^{a,*}, Marwa Qaraqe ^b, Khalid Qaraqe ^c, Erchin Serpedin ^a

^a Electrical and Computer Engineering, Texas A&M University, College Station, TX, USA

^b Information and Computing Technology, College of Science and Engineering, Hamad Bin Khalifa University, Qatar

^c Electrical and Computer Engineering, Texas A&M University at Qatar, Doha, Qatar

ARTICLE INFO

Keywords:

ECG
CNN
Transformer
Arrhythmia
Attention
Deep learning

ABSTRACT

Machine learning technologies have been applied extensively in the last decade to automatically detect and analyze various forms of arrhythmia from electrocardiogram (ECG) signals. Existing deep learning-based models focus on enhancing classification performance by exploring spatial-temporal ECG features or by implementing multi-modal and ensemble classifiers. Such approaches perform well but do not provide comprehensive accessibility for real-life applications due to the multi-lead ECG requirement. To address the issue, a single-lead ECG based network is considered. The proposed convolution, attention, and transformer-based network (CAT-Net) exhibits promising performance on arrhythmia classification by adeptly capturing local and global heartbeats' morphological characteristics. Along with the local information captured by the convolution layer, the contextual ECG information is extracted by the multi-head attention layer present in the transformer encoder. In addition, most of the existing models suffer from lower predictive performance in minority-class arrhythmias due to the highly imbalanced ECG data. To enhance the predictive performance in minority classes, three balancing techniques – SMOTE-ENN, SMOTE-Tomek, and ADASYN – are systematically evaluated, and SMOTE-Tomek is ultimately integrated. To mitigate potential dataset bias, CAT-Net was assessed across two distinct datasets: MIT-BIH and INCART, respectively, and was shown to achieve state-of-the-art performance. CAT-Net establishes new benchmarks, achieving 99.14% overall accuracy and 94.69% macro F1 score on 5-class arrhythmia classification in the MIT-BIH dataset and 99.58% accuracy with 96.15% macro F1 score for 3-class classification in the INCART dataset.

1. Introduction

Arrhythmia is a common cardiovascular disorder characterized by irregular heartbeats and manifesting in particular as slow or fast heart rates relative to a normal heart rhythm. Cardiovascular diseases (CVDs) are responsible for approximately 32% of global mortality rate leading to an annual death toll of 17.9 million individuals worldwide [1]. As a result, the early detection and classification of arrhythmias are crucial in providing early treatment and preventing adverse complications. Although arrhythmia can be detected through many means such as ECG, echocardiogram, cardiac catheterization, electrophysiology study (EPS), and stress test, ECG is the most conventional method used to detect and diagnose arrhythmia due to its simplicity and effectiveness.

Typically, a cardiologist needs to review the ECG report and the related information to diagnose the type of arrhythmia. However, the huge volume of work calls for machines to assist and even substitute cardiologists. Besides higher prediction accuracy, machines may help to reduce differences in cardiologists' interpretation. By visual assessment,

arrhythmia detection is burdensome and challenging. A case study involving 457 general practitioners (GPs) to assess the presence of atrial fibrillation (AF) reached up to 92.5% sensitivity and 89.8% specificity [2]. A well-designed machine learning (ML)-based approach can alleviate the manual task in arrhythmia detection and increase detection accuracy. This fact prompted renewed interest in exploring ML solutions.

Early ML-based approaches such as decision tree (DT) [3], support vector machine (SVM) [4], random forest (RF) [5], AdaBoost [6], and extreme gradient boosting [7] have been utilized for ECG arrhythmia classification. In addition, a large variety of approaches including but not limited to artificial neural networks (ANNs) [8], probabilistic neural networks (PNNs) [9], frequency analysis [10], path forests [11], classification and regression trees (CARTs) [12], hidden Markov models (HMMs) [13], and mixture of the expert methods [14] were proposed. In [15], ECG entropy-based features were ranked using analysis of

* Corresponding author.

E-mail address: rabiul_islam@tamu.edu (M.R. Islam).

variance (ANOVA) and the selected features were fed to the k-nearest neighbors (k-NN) and DT classifiers to detect four types of ECG beats: atrial fibrillation (A-Fib), normal sinus rhythm (NSR), ventricular fibrillation (V-Fib), and atrial flutter (AFL), respectively. The extremely unbalanced data set lowered the model's performance and a maximum accuracy of 96.3% for DT was achieved.

With the development of deep learning (DL) methods, researchers have automated the modeling process and eliminated the manual feature extraction step [16–19]. Deep learning models can be categorized into two distinct groups: (i) pure models, and (ii) hybrid models. Pure models include pure convolutional neural network (CNN)-based or pure transformer-based models while the hybrid models comprise CNN-LSTM-based models and CNN-transformer-based models. Often, the aforementioned models are updated with various types of attention mechanisms.

CNN is the simplest but most prominent DL-model [20] used for arrhythmia classification from ECG signals. Kiranyaz et al. [16] presented a one-dimensional (1D) CNN model using patient-specific long-term ECG data to classify ventricular ectopic beat (VEB) and supraventricular beat (SVEB) with 99% accuracy. In [21], a CNN model with 34 layers was employed to classify 14 types of arrhythmia with an 80% accuracy. However, this model's performance was lower than that of visual inspection by a cardiologist and it did not address ventricular flutter and fibrillation that may induce cardiac arrest.

Among the hybrid models, networks that combine CNN and recurrent neural network (RNN) architectures, such as LSTM, BiLSTM, and gated recurrent units (GRUs), have shown improved performance. For instance, Tan et al. [22] proposed a CNN and LSTM-based prototype to predict coronary artery disease (CAD) using ECG signals with an accuracy of 99.85%. Two convolutional and three LSTM layers were utilized to differentiate between normal and CAD classes. The model performed well but could not distinguish between different arrhythmia types, making it less applicable in clinics. To investigate the effect of segmentation length on performance, Oh et al. [23] presented a CNN-LSTM model that can identify NSR, paced beat (PB), ventricular premature contraction (VPC), right bundle branch block (RBBB), and left bundle branch block (LBBB). This model showed 98.1% accuracy, 97.5% sensitivity, and 98.7% specificity.

The success of attention modules in computer vision inspired researchers to implement attention-based mechanisms for ECG signal analysis [24–26]. Yao et al. proposed first a CNN-LSTM-based model [24] and then upgraded it with an attention mechanism [27] to achieve better performance. Zhang et al. [28] proposed a spatial-temporal attention-based model to classify 9 types of arrhythmia using CNN and a bidirectional GRU-based model to reach an average F1 score of 83.5%. Since 12 channels of ECG data were employed, this approach is not fit for mobile single-sensor based applications. Ullah et al. [29] proposed CNN, CNN-LSTM, and CNN-LSTM with attention modules and achieved 99.12%, 99.3% and 99.29% accuracy, respectively. However, this study ignored class balancing, and the pre-processing approach of generating model input data from the Massachusetts Institute of Technology-Beth Israel Hospital (MIT-BIH) dataset was not specified. In [30], an interpretable dual-level attentional convolutional long short-term memory neural network (DLA-CLSTM) was proposed for multilabel ECG signal classification. The model showed 88.76% and 60.14% accuracy for the MIT-BIH arrhythmia database (MITDB) and the 1st China Physiological Signal Challenge (CSCP) dataset, respectively. However, it ignored class balancing, which resulted in a reduced F1 score of 25.81% for atrial premature (AP) class. Prabhakararao and Dandapat [31] proposed inter-lead and intra-lead attention mechanisms and achieved 84.5% and 88.3% accuracy for the Physikalisch-Technische Bundesanstalt's (PTB) 12 lead PTB-XL dataset and CinC-training2017 single-lead ECG dataset, respectively. The model was implemented using a multi-scale CNN equipped with attention. However, the model neglected the global ECG characteristics. Recently proposed DSCSSA framework is a CNN

and Seq2Seq-based classifier equipped with an attention mechanism and exhibits good performance with an overall accuracy of 99.28% and macro F1 score of 95.70% for 5-class arrhythmia classification [32]. However, the DSCSSA model exploits a 2-lead MIT-BIH ECG dataset.

Recently, transformer-based models have become prominent in physiological signal processing. While CNN-LSTM models effectively capture both local and global features within ECG data, it is important to acknowledge that RNNs are confined to sequential computations. Motivated by the promising performance of transformers in NLP, researchers are currently aiming to develop pure transformer-based models as well as hybrid variants to exploit the parallel computing features of transformers. Hu et al. [33] proposed an ECG detection transformer (DETR) for continuous ECG segments to classify 8, 4, and 2 distinct types of arrhythmia. Yan et al. [34] integrated manually constructed features with transformer encoder-generated features to enhance the predictive performance for four distinct categories of ECG heartbeats. To reduce the number of parameters in the transformer model, Meng et al. [35] replaced self-attention of the Fusing Transformer with the LightConv Attention (LCA) and obtained 99.32% accuracy. Le utilized 1D ECG, 2D ECG spectrogram, and age and gender metadata to capture time series, visual-temporal, and metadata-based information, respectively. They presented a multi-mode RCNN with a transformer encoder to categorize arrhythmias by fusing features. The model is overly complex, with an accuracy of only 98.29%. Che et al. [36] evaluated CNN-Transformer-based hybrid models against CNN and CNN-BiLSTM models with and without attention, and found that the CNN-Transformer model performed best. They improved efficiency by adding a link constraint to the loss function; however, the model required 12 leads ECG data, making it difficult for practical implementation.

With the advancement of technology, the development of wearable single-lead ECG devices is transforming continuous ECG recordings, with the ability to record and transmit ECG signals to smartphones, computers, and IoT systems for analysis. AliveCor KardiaMobile EKG, KardiaMobile® Card, EMAY Portable ECG Monitor, and Eko DUO ECG are a few examples. The analyzing models related to these devices can detect an abnormal cardiac rhythm but face difficulties in classifying various arrhythmias. Consequently, a model with higher accuracy and compatibility with single-lead ECG devices is a fundamental prerequisite for real-life applications. Limited work has been reported on single-lead ECG based arrhythmia detection. The model of [37] showed only 87.8% accuracy, while the model proposed in [38] achieved accuracy levels of 93.63% and 95.87% for SVEB and VEB classes, respectively.

To enable adoption in practice of heart arrhythmia detection and classification applications via single-lead ECG devices and smart devices, we propose a CNN-Transformer-based hybrid network (CAT-Net) for the detection and classification of heart arrhythmia. CNN computes ECG local features, channel attention extracts the most relevant portion, and a transformer contextualizes the ECG feature map in latent space. The incorporation of channel attention after CNN and transformer on the ECG feature map enabled to focus on local, significant, and global contextualization, making our model superior to the existing state-of-the-art models. The proposed model is shown to classify five categories of arrhythmia with 99.14% accuracy and 94.69% macro F1 score. In addition, the proposed model outperforms existing single-lead ECG-based models in the literature. The model presents significant impact on analyzing the presence of five types of arrhythmia in ECG recordings.

Fig. 1 illustrates the high-level system architecture of the proposed single-lead arrhythmia detection model. The main novel aspects of this work are next summarized.

- (1) **Novel Model:** We present a novel CAT-Net model that employs convolution and attention for capturing local insights and a transformer encoder for modeling global heartbeat dependencies. The transformer encoder contributes to contextualizing ECG feature maps, enabling parallel processing unlike LSTM.

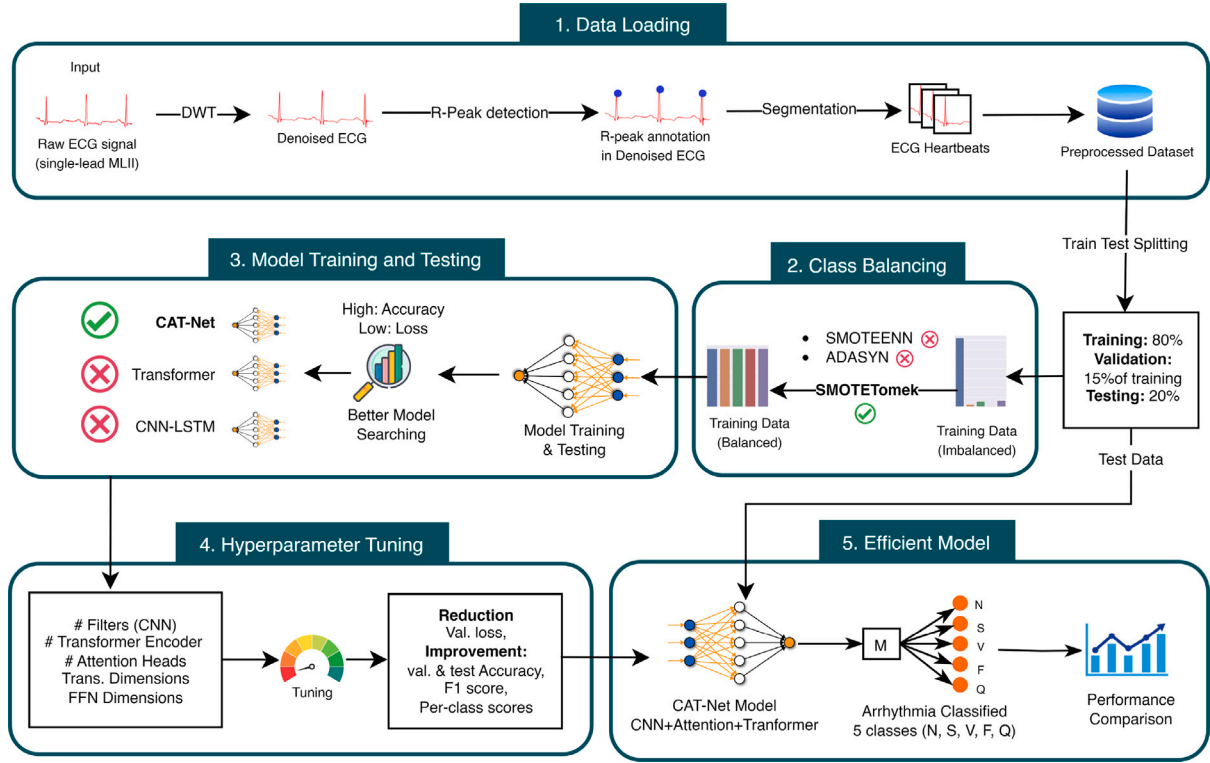


Fig. 1. Overview of the proposed processing pipeline.

- (2) **Dataset Invariance:** The proposed CAT-Net model is evaluated with two different datasets and it generated state-of-the-art results for both datasets. Therefore, the model is generalized and dataset invariant.
- (3) **Focusing Minority Class:** To emphasize the minority class's predictive performance, 3 class balancing approaches: Synthetic Minority Oversampling Technique Edited Nearest Neighbor (SMOTE-ENN), SMOTE-Tomek, and Adaptive Synthetic (ADASYN) were assessed. By removing Tomek links, SMOTE-Tomek outperforms and is finally integrated into the proposed model.
- (4) **Usability:** The proposed model requires only 1-lead ECG data, making it usable in automatic, wearable, and IoT-based real-world applications.
- (5) **Performance:** A comprehensive empirical assessment is performed by systematically manipulating the model configurations and hyperparameters of CNN, channel attention, and transformer encoder layers. Finally, an effective model with higher accuracy (of 99.14% and 99.58% for two distinct datasets) is developed.

The developed model is anticipated to support the following applications: (i) early arrhythmia detection, (ii) real-time arrhythmia classification, and (iii) patient and physician notification via single-lead ECG wearable devices including smart shirts and other recording devices.

The rest of paper is organized as follows. Section 2 formulates the problem, describes the data set, and outlines the relevant preprocessing steps and proposed modeling framework. The model's performance is analyzed in Section 3. Section 4 concludes the paper.

2. Methodology

This section presents an overview of the used datasets, data preprocessing techniques, and class balancing methodologies. Afterward, the proposed CAT-Net model architecture is introduced, comprising

three distinct components: convolution, attention, and transformer. This section also presents Python implementation details.

2.1. Problem formulation

The objective of this study is to maximize the predictive performance of 5 classes of arrhythmia classification with minimum lead ECG data. The input dataset representing the set of heartbeats is given by $X = \{x_1, x_2, x_3, \dots, x_N\}$, where N denotes the number of heartbeats. The variable $x_i = (s_1, s_2, s_3, \dots, s_n)$ collects the ECG signal values and n stands for the length of the segmentation window. The Association for the Advancement of Medical Instrumentation (AAMI) [39] recommends classifying MIT-BIH dataset arrhythmia as normal beat (N), supraventricular ectopic beat (S), ventricular ectopic beat (V), fusion beat (F), and unknown beat (Q). Accordingly, the output is expressed as $Y = \{y_1, y_2, y_3, \dots, y_N\}$, where

$$y_i = \begin{cases} 0, & N \text{ class} \\ 1, & S \text{ class} \\ 2, & V \text{ class} \\ 3, & F \text{ class} \\ 4, & Q \text{ class} \end{cases}$$

This study aims to find an effective model f such that $f : X \rightarrow Y$ predicts the arrhythmia type for a given ECG heartbeat with high accuracy.

2.2. Datasets

This study employed two datasets, (i) MIT-BIH Arrhythmia Database [40] and (ii) St. Petersburg INCART 12-lead Arrhythmia Database [41]. Table 1 summarizes information about the datasets while the number of heartbeat counts for each of the relevant categories is presented in Table 2.

Table 1
Utilized ECG dataset information.

Topic	MIT-BIH (Dataset 1)	INCART (Dataset 2)
Number of Recordings	48	75
Laboratory	Beth Israel Hospital Arrhythmia Laboratory	St. Petersburg Institute of Cardiological Technics (INCART)
Acquisition tool	Holter recordings	Holter recordings
Signals	2 Lead ECG, (i) MLII and (ii) V1; (occasionally V2, V4 or V5)	12 leads (Only lead II signals are used)
Signal length, sampling frequency	30 min (or slightly over), 360 Hz	30 min, 257 Hz
Subjects	47 (25 Men, 22 Women). Only 1 patient has 2 recordings.	32 (17 Men, 15 Women)
Age Range	Men: (32~89) years, Women: (23~89) years	18~80 years
Number of Heartbeat Types	20 (Table 2 lists 15 types of heartbeats are used)	11 (6 types of heartbeats are used)
Recording Period	1975~1979	(Not Available)

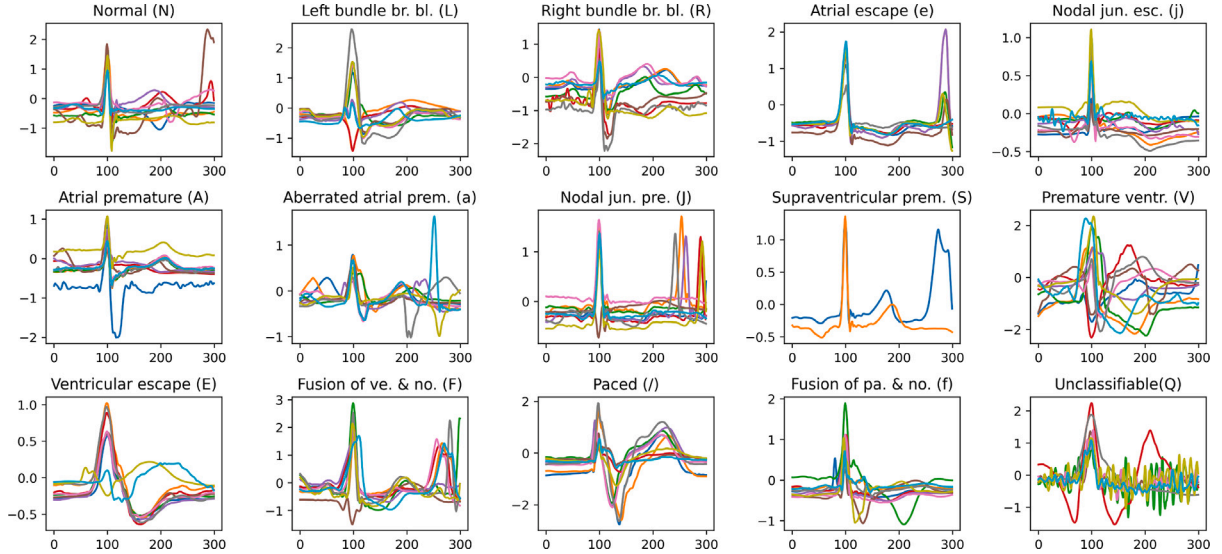


Fig. 2. Visualization of 15 different types of ECG heartbeats of MIT-BIH dataset. Each type consists of 10 randomly selected heartbeat samples except supraventricular premature beat (S) which assumes only 2 beats in the data set. Different colors are used to differentiate heartbeats.

2.2.1. MIT-BIH arrhythmia database

The MIT-BIH dataset [40] comprises 48 samples of 2-lead ECG recordings, each lasting around 30 min. These recordings were collected between 1975 and 1979 through a collaboration between MIT and Boston's Beth Israel Hospital. It contains 47 ECG recordings from patients of both genders (25 men and 22 women), covering 32–89 years of age (men) and 23–89 years of age (women). From a set of 4000 24 hour Holter recordings of a mixed population (inpatients 60%, outpatients 40%) of BIH, 23 samples were chosen randomly, while the remaining 25 samples were selected that had less common but clinically significant arrhythmias. In this study, only modified limb lead II (MLII) signals are used to build a single-lead ECG model. Therefore, 46 patients' signals were used because the remaining two patients' ECG records only have lead V2, V5, or V5 signals, not lead MLII. Although the original MIT-BIH data consists of 23 distinct categories of heartbeat, only 15 types stated in Table 2 are classified into five major classes: N, S, V, F, and Q. Fig. 2 depicts the morphological aspects of 15 types of heartbeats.

2.2.2. St. Petersburg INCART 12-lead arrhythmia database

The INCART dataset [41] consists of 75 annotated recordings extracted from 32 Holter records. The signals were collected from the patients (17 men and 15 women, aged 18–80; mean age: 58) undergoing tests for coronary artery disease. There were no patients with pacemakers; the majority experienced ventricular ectopic beats. In total, the reference annotation files comprise more than 175,000 beat annotations. Among the 12 leads, only lead II data was extracted for our model. The dataset is utilized to categorize 3 types of arrhythmias because fusion beat (F) and unknown beat (Q) have only 219 and 6 heartbeats, respectively, as mentioned in Table 2. The sample heartbeats of 3 categories and 6 sub-categories are visualized in Fig. 3.

Using the MIT-BIH dataset, we first develop our model for 5-class classification and then evaluate it for hyperparameter adjustments, class balancing, and architecture changing. The INCART dataset is then used to validate our model's performance and check its robustness.

2.3. Data preprocessing

Data preprocessing assumes two consecutive steps: (i) signal denoising and (ii) heartbeat segmentation. The raw ECG signals downloaded from the MIT-BIH and INCART website are perturbed by noise induced by the power line interface, myoelectric interface, etc. The popular discrete wavelet transform (DWT)-based method is used to denoise the 1D ECG signal. The wavelet denoising procedure undertakes three distinct operations to remove noise from the ECG signal: (i) wavelet decomposition, (ii) coefficient processing, and (iii) wavelet reconstruction, respectively.

The raw ECG signal is first decomposed into nine levels by employing Daubechies wavelet 5 (db5), and 9 Details coefficients (cD1, cD2, cD3, cD4, cD5, cD6, cD7, cD8 and cD9) and 1 Approximate coefficient (cA9). Then, two detail coefficients, cD1 and cD2, are turned into zero and the remaining 7 detail coefficients (cD3 to cD9) are processed using the 'soft thresholding' method. Finally, the processed approximate and detail coefficients are used to reconstruct the denoised ECG signal using the wavelet reconstruction technique. Fig. 4 shows the ECG signals before and after denoising.

After denoising, the R-peak position is used for signal segmentation. There are many R-peak detection techniques and segmentation algorithms for ECG signals proposed in the literature, including but not limited to ANN, genetic algorithm, wavelet transform, and modified Pan-Tompkins algorithm [42]. The location of the R peak is already

Table 2
Heartbeats count of MIT-BIH and INCART datasets.

Classes	SN	Type	Heartbeat name	# Heartbeats MIT-BIH (INCART)	# Sub-total heartbeats MIT-BIH (INCART)
Normal (N)	1	N	Normal beat	74658 (150410)	90210 (153676)
	2	L	Left bundle branch block beat	8063	
	3	R	Right bundle branch block beat	7244 (3174)	
	4	e	Atrial escape beat	16	
	5	j	Nodal (junctional) escape beat	229 (92)	
Supraventricular ectopic beat (S)	6	A	Atrial premature beat	2540 (1944)	2775 (1960)
	7	a	Aberrated atrial premature beat	150	
	8	J	Nodal (junctional) premature beat	83	
	9	S	Supraventricular premature beat	2 (16)	
Ventricular ectopic beat (V)	10	V	Premature ventricular contraction	7117 (20013)	7223 (20013)
	11	E	Ventricular escape beat	106	
Fusion Beat (F)	12	F	Fusion of ventricular and normal beat	802 (219)	802 (219)
Unknown beat (Q)	13	/	Paced beat	3612	3887 (6)
	14	f	Fusion of paced and normal beat	260	
	15	Q	Unclassifiable beat	15 (6)	
				# Total heartbeats	104897 (175874)

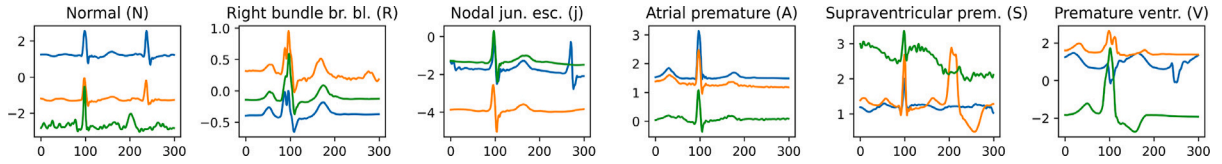


Fig. 3. Visualization of 6 different types of ECG heartbeats of INCART dataset. Each type consists of 3 randomly selected heartbeat samples. Different colors are used to differentiate heartbeats.

annotated in the MIT-BIH dataset. Therefore, the annotated R-peak locations and the fixed window method for segmentation are used. A window of size 300 samples is used where 99 samples are taken from the left and 201 samples are taken from the right of the R-peak position.

2.4. Class balancing

The MIT-BIH and INCART datasets are subject to an imbalanced number of samples (as shown in Table 2) which biases machine learning (ML) algorithms to learn attributes better for the larger class dataset and to perform poorly in predicting the minority class data. To balance the datasets we applied three class balancing approaches: SMOTE-ENN, SMOTE-Tomek, and ADASYN. SMOTE-Tomek is employed in our model for its superior performance. SMOTE-ENN and SMOTE-Tomek were introduced by Batista et al. [43] which offer improvements to the SMOTE algorithm [44].

In SMOTE, new synthetic data are generated rather than oversampling with replacement. First, the difference between a minority class sample x_i and one of its k nearest neighbors $\hat{x}_i$ is multiplied by a factor δ . This product is added to the selected samples x_i to create a new sample x_{new} as follows

$$x_{new} = x_i + (\hat{x}_i - x_i) \times \delta$$

where δ ranges from 0 to 1. SMOTE-Tomek is a combination of oversampling by SMOTE and undersampling by removing Tomek links. Tomek Links is a method for detecting pairs of nearest neighbors with various classes in a dataset. A pair of minimally Euclidian distance neighbors (x_i, x_j) with x_i belonging to the minority class and x_j to

the majority class is said to be Tomek Links if there is no sample x_k that satisfies the condition: $d(x_i, x_k) < d(x_i, x_j)$ or $d(x_j, x_k) < d(x_j, x_i)$, where $d(x_i, x_j)$ represents the Euclidian distances between x_i and x_j . ADASYN [45] mainly used density distribution to decide the number of synthetic samples s_i to be generated, following the equation as follows

$$s_i = x_i + (x_{zi} - x_i) \times \lambda$$

where $(x_{zi} - x_i)$ is the difference vector in n-dimensional spaces, and λ is a random number: $\lambda \in [0, 1]$.

SMOTE-Tomek offers a dual advantage. First, while SMOTE may introduce synthetic noisy instances, Tomek link removal effectively eliminates such instances, ensuring a cleaner and more reliable dataset for training. Second, SMOTE-Tomek simultaneously addresses both oversampling and undersampling challenges, resulting in a more balanced dataset. This balanced distribution enhances the classifier's ability to generalize effectively across all classes. In contrast, SMOTE-ENN and ADASYN did not fully exploit the dual benefits of oversampling and undersampling simultaneously, leading to slightly reduced classification performance in arrhythmia classification from ECG.

The number of training and testing samples and corresponding class balancing approaches for MIT-BIH and INCART datasets are provided in Tables 3 and 4, respectively. For both datasets, as the 'N' class avails a larger number of samples, random under-sampling (RUS) is applied before employing the SMOTE-Tomek approach. Finally, two balanced datasets with 55k and 44k samples per class are generated.

2.5. Model architecture

The classification model architecture consists of three main components: (i) CNN, (ii) attention mechanism, and (iii) transformer encoder.

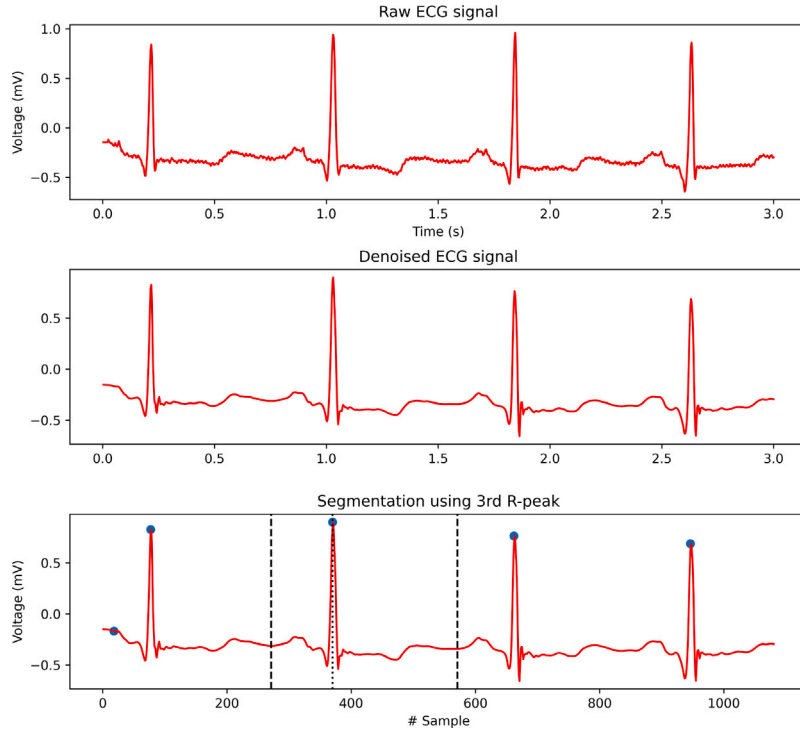


Fig. 4. Raw, denoised and R peak detected ECG signal from MIT-BIH dataset (recording number 100, channel MLIJ). The third graph shows segmentation of window size 300, with the third R-peak (location 370) as a reference point.

Table 3

Number of samples in training, testing, and class balancing approach: MIT-BIH dataset.

Class	# Heartbeats	# Testing (20%)	# Training	Balancing technique(s)	# Training (Balanced)
N	90210	17982	72228	RUS + SMOTE-Tomek	55000
S	2775	568	2207	SMOTE-Tomek	55000
V	7223	1463	5760	SMOTE-Tomek	55000
F	802	167	635	SMOTE-Tomek	55000
Q	3887	800	3087	SMOTE-Tomek	55000
Total	104897	20980	83917	–	275000

Table 4

Number of samples in training, testing, and class balancing approach: INCART dataset.

Class	# Heartbeats Used	# Testing (20%)	# Training	Balancing technique(s)	# Training (Balanced)
N	153343	30624	122719	RUS + SMOTE-Tomek	40000
S	1957	413	1544	SMOTE-Tomek	40000
V	19980	4019	15961	SMOTE-Tomek	40000
Total	175280	35056	140224	–	120000

The transformer encoder is described in a separate sub-section with its components. At the end of the complete framework, the feature map is sent to a flattened layer and to the dense layers for prediction. The proposed CAT-Net model is illustrated in Fig. 5.

2.5.1. CNN architecture

The successful use of CNNs in computer vision and image processing applications was followed up by outstanding results in the analysis of time series data such as ECG and EEG. Many challenging problems have already been addressed by employing 1D CNN models. The convolution operation allows the consideration of local information from a signal to create a feature map for the next layer.

The considered model employs four CNN layers. A summary of the model framework is presented in Table 5. The first CNN layer processes the input ECG and the next layer considers the previous layer's feature map as input. For the first convolutional layer (CL), 16 filters of size 21×1 with stride 1 are applied with the same padding. Next, after the attention layer, a 3×1 sized 1D max pooling layer with stride

2 is applied. Later, for the 2nd, 3rd and 4th CLs, 32 filters of size 23×1 , 64 filters of size 25×1 , and 128 filters of size 27×1 are employed, respectively. Each layer maintains the 'same' padding with a stride value of 1. The rectified linear unit (ReLU) activation function is followed by each CL.

The number of filters and the kernel size are gradually increased with the depth of network since the deeper layers consider more information for prediction. To reduce the network computational complexity, the pooling layer is employed after the first 3 CLs. Following the first two CLs, a 1D maximum pooling of size 3×1 , stride 2, and 'same' padding is applied. After the 3rd CL, a 1D average pooling of size 3×1 , stride 2, and 'same' padding is employed. As the deeper portion contains sensitive information, average pooling is applied instead of maximum pooling like the previous two layers.

For the convolutional layers CL1, CL2, CL3 and CL4, the conducted operations are

$$x_k^l = \sigma \left(\sum_{i \in M_k} x_i^{l-1} * \omega_{ik} + b_k \right), \quad (1)$$

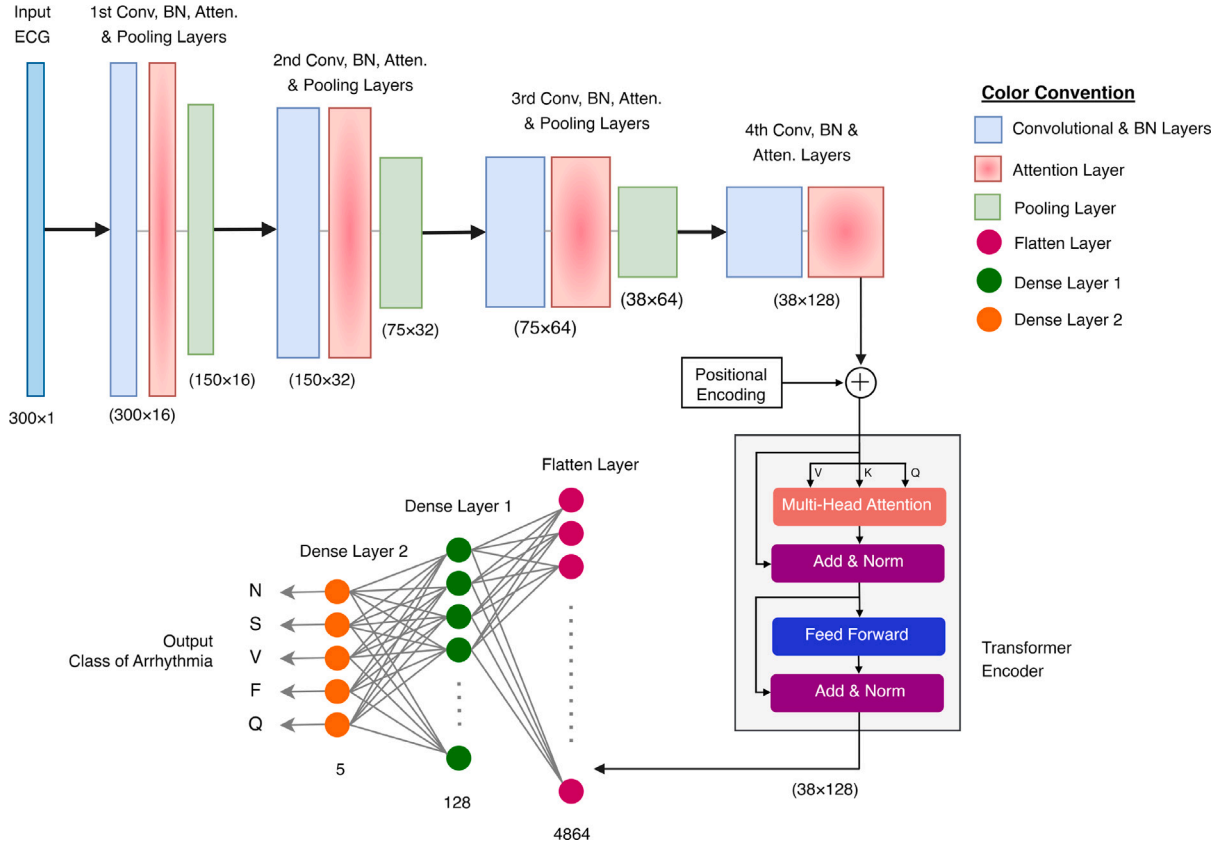


Fig. 5. Proposed CAT-Net model for arrhythmia classification from ECG signal. The model predicts five types of arrhythmia from ECG heartbeats: normal (N), supraventricular ectopic (S), ventricular ectopic (V), fusion (F), and unknown (Q).

Table 5
Summary of the proposed CAT-Net model.

Layer	# Filters	Kernel size	Output shape	# Parameters
Conv1D	16	21×1	300×16	352
BatchNorm	–	–	300×16	64
Attention1D	16	–	300×16	0
MaxPooling1D	–	3×1	150×16	0
Conv1D	32	23×1	150×32	11808
BatchNorm	–	–	150×32	128
Attention1D	32	–	150×32	0
MaxPooling1D	–	3×1	75×32	0
Conv1D	64	25×1	75×64	51264
BatchNorm	–	–	75×64	256
Attention1D	64	–	75×64	0
AveragePooling1D	–	3×1	38×64	0
Conv1D	128	27×1	38×128	221312
BatchNorm	–	–	38×128	512
Attention1D	128	–	38×128	0
TransformerEncoder	128 dim	4 heads	38×128	280576
Flatten	–	–	4864	0
Dense	–	–	128	622720
Dropout	–	–	128	0
Dense	–	–	5	645

where x_k^l and x_k^{l-1} represent the k th neuron output at layers l and $l-1$, respectively. The weight kernel between the i th and k th neuron is denoted by ω_{ik} and the bias of k th neuron is indicated by b_k . Sigma (σ) denotes the ReLU activation function. Notation M_k stands for the active range of the convolutional kernel.

After each convolution layer, a batch normalization (BN) layer is applied to preserve the mean output close to 0 and the output standard deviation close to 1. The BN layer contributes to stabilizing the learning process and reducing the number of training epochs required to train deep networks.

2.5.2. Attention mechanism

In the proposed hybrid model, each CL is followed by a channel attention mechanism. The core idea of applying attention mechanism is to enhance the training performance by emphasizing the more informative temporal segment of ECG signal. As only single-lead ECG data is used, the feature map becomes a 2D matrix and therefore, only channel attention is implemented following the CBAM strategy [46]. The proposed attention mechanism schematic is presented in Fig. 6.

The 2D feature map after each convolution is represented by matrix $F \in \mathbb{R}^{C \times L}$, where C indicates the number of channels and L denotes the length. In the channel attention mechanism, the 2D feature map is first converted into the 1D attention map $A_c \in \mathbb{R}^{C \times 1}$. Then, the 1D attention map A_c is multiplied with the feature map F^l to generate the attention-based refined feature map F^{l+1} for the next layer as illustrated in Fig. 6. The overall process is formalized at layer l through the element-wise matrix multiplication operator $\otimes$ as follows

$$F^{l+1} = A_c(F^l) \otimes F^l. \quad (2)$$

Feature map of l th layer is represented by F^l .

To compute the channel attention, the spatial dimension of the input feature is squeezed. The spatial information is aggregated by using max-pooling and average pooling in the spatial domain. Thus, two different spatial presenters are generated F_{avg}^c and F_{max}^c , respectively. Both presenters are then inputted in a shared multi-layer perceptron (MLP) with only one hidden layer. Next, element-wise summation is applied to the generated output feature vectors. The attention map is formally generated as follows

$$\begin{aligned} A_c(F) &= \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \\ &= \sigma \left(W_1 \left(W_0 \left(F_{avg}^c \right) \right) + W_1 \left(W_0 \left(F_{max}^c \right) \right) \right). \end{aligned} \quad (3)$$

The weights of MLP are incorporated in matrices $W_0 \in \mathbb{R}^{r \times C}$ and $W_1 \in \mathbb{R}^{C \times r}$. Variable r is introduced to reduce the parameter overhead.

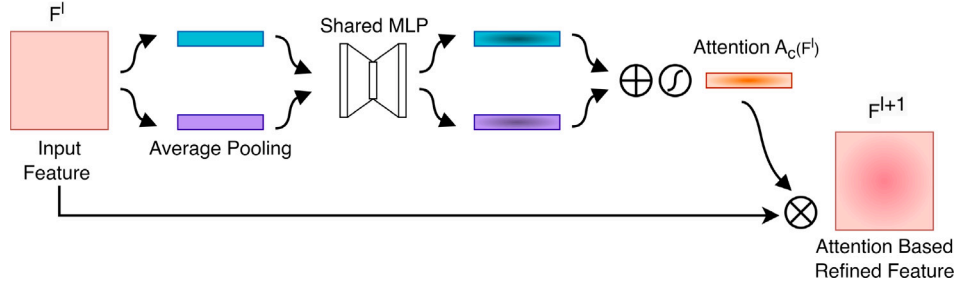


Fig. 6. Channel attention mechanism along the process of refining the feature map.

For our network, $r = 8$. Thus, the hidden activation size is reshaped to $\mathbb{R}^{C \times 1 \times 1}$.

The 4th attention layer's output feeds into the subsequent transformer encoder layer described next. Later, the transformer encoder's output of size 38×128 is flattened into a 4864-sized vector. Finally, this vector is processed through a dense layer of 128 neurons, followed by another dense layer of 5 neurons using the 'softmax' activation function to classify 5 types of arrhythmia.

2.6. Transformer encoder

The CNN and attention layers model the local morphological information of ECG. To capture long-term ECG signal dependencies, a transformer encoder is utilized. The transformer is a groundbreaking Seq2Seq architecture applied in NLP [47]. To adapt the transformer in arrhythmia classification from ECG, the decoder part is eliminated. A transformer encoder includes positional encoding, a multi-head attention layer, and a feed-forward layer along with skip connections and normalization layers.

2.6.1. Positional encoding (PE)

As the transformer does not include recurrence or convolution, understanding the order of sequence positional encoding is important. Therefore, before entering the transformer encoder the input embedding (i.e. ECG feature map) is added with PE. The PE must have the same dimensions as the input embedding. Among different approaches, sine and cosine functions of different frequencies are used for PE as they work efficiently compared to other approaches. The PE is computed as follows

$$PE_{(pos, 2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos, 2i+1)} = \cos(pos/10000^{2i/d_{model}}),$$

where pos identifies the position, i denotes the dimension ranging from 0 to $d_{model}/2$, and d_{model} is the model dimension and refers to the dimension of the input token (i.e., segmented ECG feature map). The main function of PE is to help the model in learning relative positions easily. The PE is computed once during training and can be used subsequently.

2.6.2. Multi-head attention

To empower the ECG feature map by contextualizing the long-range dependencies, a multi-head attention (MHA) layer is implemented as the first layer of transformer encoder. The internal computation of MHA internal is similar to scaled dot-product attention. The difference is that in MHA, scaled dot-product attention is performed in parallel for different heads. In our model, we implemented 4 heads. The computation process of scaled dot-product and multi-head attention layers is illustrated in Fig. 7.

First, the positional encoded ECG feature map $F_{ecg} \in \mathbb{R}^{L_{ecg} \times d_{model}}$ is mapped into three matrices: query $Q \in \mathbb{R}^{L_{ecg} \times d_q}$, key $K \in \mathbb{R}^{L_{ecg} \times d_k}$, and values $V \in \mathbb{R}^{L_{ecg} \times d_v}$. Here, L_{ecg} denotes the length of ECG feature map. Next, the dot-product of query and key is scaled with factor $1/\sqrt{d_k}$ for

getting balanced gradients after the softmax function. The dot-product attention is calculated as follows

$$Attention(Q, K, V) = Softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V.$$

For MHA, the query, key, and value matrices are linearly projected as $W_i^Q \in \mathbb{R}^{d_{model} \times d_q}$, $W_i^K \in \mathbb{R}^{d_{model} \times d_k}$, and $W_i^V \in \mathbb{R}^{d_{model} \times d_v}$, respectively. Next, the outputs of all heads are concatenated and fed to a linear layer to produce the output:

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h),$$

where $head_i = Attention(QW_i^Q, KW_i^K, VW_i^V)$.

The MHA output (Z_{mha}) is added and normalized to generate the final output (Z) and fed to the next feed-forward layer.

$$Z = LayerNorm(Z_{mha} + F_{ecg}).$$

2.6.3. Feed forward network

The output Z is fed to the next sublayer named feed-forward network (FFN). It consists of two linear transformations with a ReLU activation function in between. The output of FFN layer (Z_{ffn}) is given by

$$Z_{ffn} = FFN(Z) = \max(0, ZW_1 + b_1)W_2 + b_2,$$

where W_1 , W_2 denote the weights and b_1 , b_2 represent the biases. Finally, the Z_{ffn} is transferred to the 'Add and Norm' layer to generate the transformer encoder's final output (Z_{TE})

$$Z_{TE} = LayerNorm(Z_{ffn} + Z).$$

The skip connections always keep the previous layer's impact, which helps predicting with higher performance.

2.7. Computational environment

The model is implemented in 'Google Colab' using Python (version 3.8.16). The 'wfdb' (waveform database) and 'pywt' (PyWavelets) library packages were used for ECG signal processing and denoising, respectively. TensorFlow and Scikit-learn libraries were utilized to develop and evaluate the model. The CAT-Net model was developed using Keras with TensorFlow backend. Regarding the computational infrastructure, the study utilized Google Colab's GPU, featuring an Intel® Xeon® CPU @2.00 GHz, 64-bit, GPU Memory: 12 GB/16 GB and GPU Memory Clock: 0.82 GHz/1.59 GHz.

2.8. Training phase

The overall preprocessed data is split into 80% training and 20% testing. The CAT-Net model is trained with mini-batch gradient descent with a batch size of 128. The model is trained with 'adam' optimizer for 60 epochs. Only 15% of training data is considered as validation data.

For every training step, the model weights are recorded, and the best weights are saved using 'modelcheckpoint' callback function. The validation accuracy is set as the indicator for searching the best model.

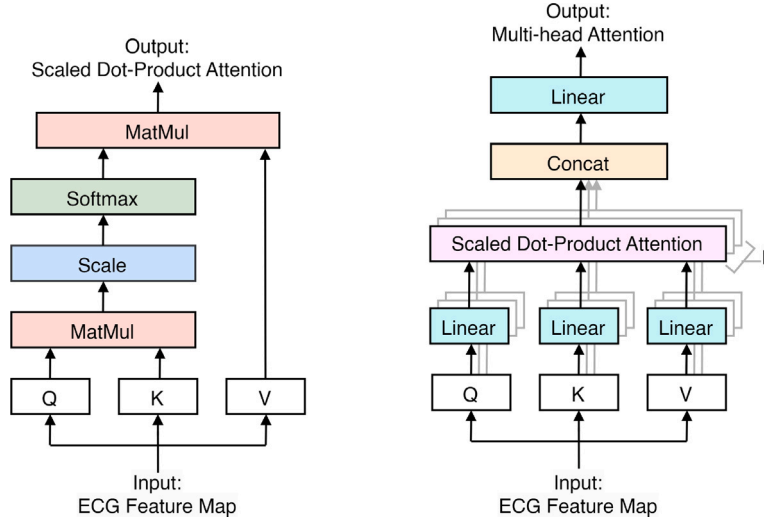


Fig. 7. The computation flow in scaled dot-product attention-(left) and multi-head attention-(right).

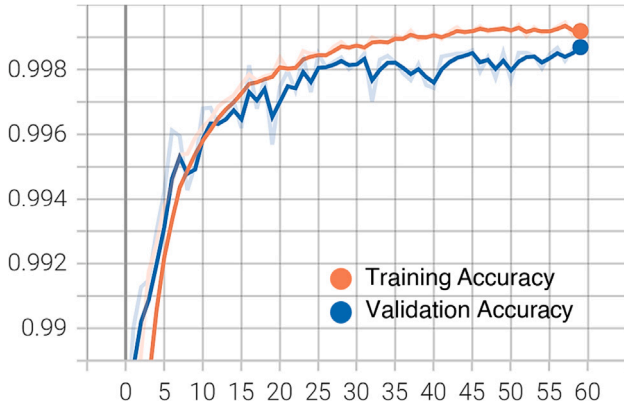


Fig. 8. Accuracy curves during training the model. Curves are plotted in 'Tensorboard'. The original shaded curves are smoothed by a factor of 0.5.

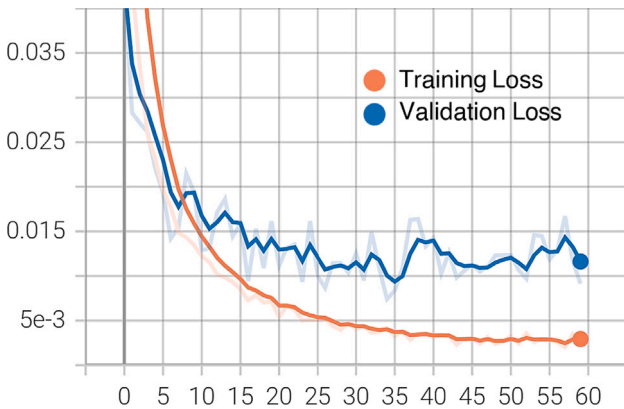


Fig. 9. Loss curves during training the model. Curves are plotted in 'Tensorboard'. The original shaded curves are smoothed by a factor of 0.5.

The 'tensorboard' callback is also used to check the model progress during training. The training and testing accuracy and loss curves during training are presented in Figs. 8 and 9, respectively.

The model architecture and hyperparameters are varied following all the possible paths in Fig. 10. After designing a simple CNN-based

Table 6

The values of model hyperparameters.

Hyperparameter	Value	Notation
Batch size	128	–
Epochs	60	–
Model dimensions	128	d_{model}
Attention heads	4	h
FNN dimensions	128	d_{FNN}
Dropout rate	0.02	–

model, the LSTM, attention, and transformer encoder layers are incorporated gradually. The performance of the model comprising four consecutive CNN layers having 4, 16, 32, 64, and 128 filters is evaluated. Random search methods are employed to obtain a better model. Next, the hyperparameters of transformer encoder are set carefully. Finally, based on accuracy and macro F1 score the CAT-Net model architecture is confirmed. The values of hyperparameters used for the whole model and transformer encoder are presented in Table 6.

3. Results and discussion

Next, the model performance based on different evaluation matrices is assessed and analyzed. The effect of changing hyperparameters, class balancing, and small data in 'F' class and the model's robustness are discussed. The proposed work is benchmarked with state-of-the-art approaches and future research recommendations are put forward.

3.1. Evaluation matrices

The classification model performance is assessed via accuracy, sensitivity, specificity, and F1 score, which is an important factor to evaluate the performance of a machine learning model. The per-class accuracy, sensitivity, and specificity are computed according to (4) to (7).

$$Sensitivity_{PC} = Sensitivity_{OA} = \frac{TP}{TP + FN} \quad (4)$$

$$Specificity_{PC} = Specificity_{OA} = \frac{TN}{TN + FP} \quad (5)$$

$$Accuracy_{PC} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

$$Accuracy_{OA} = \frac{N_{correct}}{N}, \quad (7)$$

where TP , TN , FP , and FN refer to true positive, true negative, false positive, and false negative, respectively; OA indicates overall performance, PC denotes per-class performance, N stands for the number of

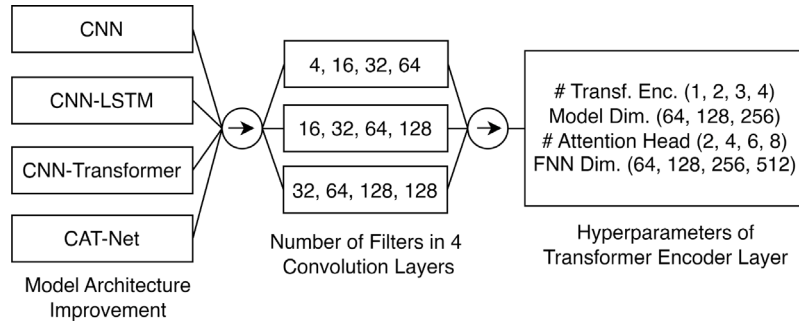


Fig. 10. Variation of model and related hyperparameters.

Table 7

Confusion matrix and per-class performance score of CAT-Net model on MIT-BIH testing data (Imbalanced)

	Class type	Predicted class					Per-class score				Total data per-class
		N	S	V	F	Q	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	
True class	N	17910	44	15	11	2	99.27	99.60	97.26	99.57	17982
	S	37	531	0	0	0	99.60	93.49	99.77	92.75	568
	V	23	2	1422	16	0	99.69	97.19	99.88	97.80	1463
	F	21	0	7	139	0	99.74	83.23	99.87	83.48	167
	Q	1	0	1	0	798	99.98	99.75	99.99	99.75	800
Overall Performance							99.14	99.14	99.14	99.14 Micro 94.69 Macro 99.14 Weighted	20980

total samples and $N_{correct}$ identifies the number of samples predicted correctly.

The per-class F1 score is calculated as follows

$$F_{1,PC} = \text{Micro } F_{1,OA}$$

$$= \frac{2 \times \text{precision} \times \text{sensitivity}}{\text{precision} + \text{sensitivity}} \quad (8)$$

$$= \frac{2TP}{2TP + FP + FN},$$

where $F_{1,OA}$ indicates overall F1 score.

The overall macro and weighted F1 score are computed as follows

$$\text{Macro } F_{1,OA} = \frac{1}{C} \sum_{i=1}^C F_{1,PC} \quad (9)$$

and

$$\text{Weighted } F_{1,OA} = \frac{1}{N} \sum_{i=1}^C F_{1,PC} \times N_i, \quad (10)$$

where C denotes the number of classes, $F_{1,PC}$ indicates the F1 score of i th class, and N_i represents the number of samples in i th class. For multi-class classification, the overall model's sensitivity, specificity, accuracy, and Micro F1 are equal as

$$\begin{aligned} \text{Sensitivity}_{OA} &= \text{Specificity}_{OA} \\ &= \text{Accuracy}_{OA} \\ &= \text{Micro } F_{1,OA}. \end{aligned} \quad (11)$$

3.2. Classification performance

The confusion matrix and the per-class classification report MIT-BIH data are presented in Table 7. The confusion matrix shows that 44 'N' beats are mispredicted as 'S' and 37 of S-type beats are wrongly predicted as 'N'. The model falls in difficulty in discriminating between normal and supraventricular beats. From the heartbeat plotting in Fig. 2, the reason is obvious as these two types of heartbeats contain almost the same morphological pattern. Due to distinguishable variations in unclassifiable heartbeats (Q), the accuracy and F1 scores are excellent in the class. Among 800 beats, only 2 beats are misclassified. The overall accuracy of the proposed system is 99.14%.

It is observed that the per-class accuracy is always greater than 99.27% as stated in Table 7. This demonstrates the model's outstanding performance across all the classes. When the model was run without class balancing for the training data, it showed good performance for the N class but very poor performance in the F class due to the shortage of F class data sets. The model presented 99.60%, 93.49%, 97.19%, 83.23% and 99.75% sensitivity for N, S, V, F and Q, respectively. The sensitivity of 'F' class is yet lower compared to the other classes. Thus, improving the F class sensitivity represents an open research problem.

The proposed CAT-Net model achieves 94.69% macro and 99.14% weighted F1 scores. It is noticed that for the S and F classes, sensitivity and F1 scores are lower compared to the other three classes. This occurs due to less number of samples in the original data for these two classes.

3.3. Adjustment of model hyperparameters

Using a random search approach, the model architecture and its hyperparameters, such as the number of filters in CNN, the number of transformer encoders and other parameters are adjusted. In every step, the accuracy, number of total parameters, training time, validation accuracy and loss, etc., are recorded. Table 8 provides a listing of notable models and their corresponding performance. From Table 8, it is clear that the CNN-Net model offers the highest validation accuracy (99.89%) and the lowest validation loss (0.52%). Even though the number of parameters and training time per epoch is slightly higher than in other models, it could be ignored for better results.

The developed model is trained and tested with different transformer encoder hyperparameters to explore their effects. Fig. 11 depicts the corresponding accuracy and macro F1 score. After investigation, we found that only 1 encoder with 4 attention heads for MHA layer works effectively for ECG arrhythmia classification. We set the model and FNN dimension to 128 for the highest performance.

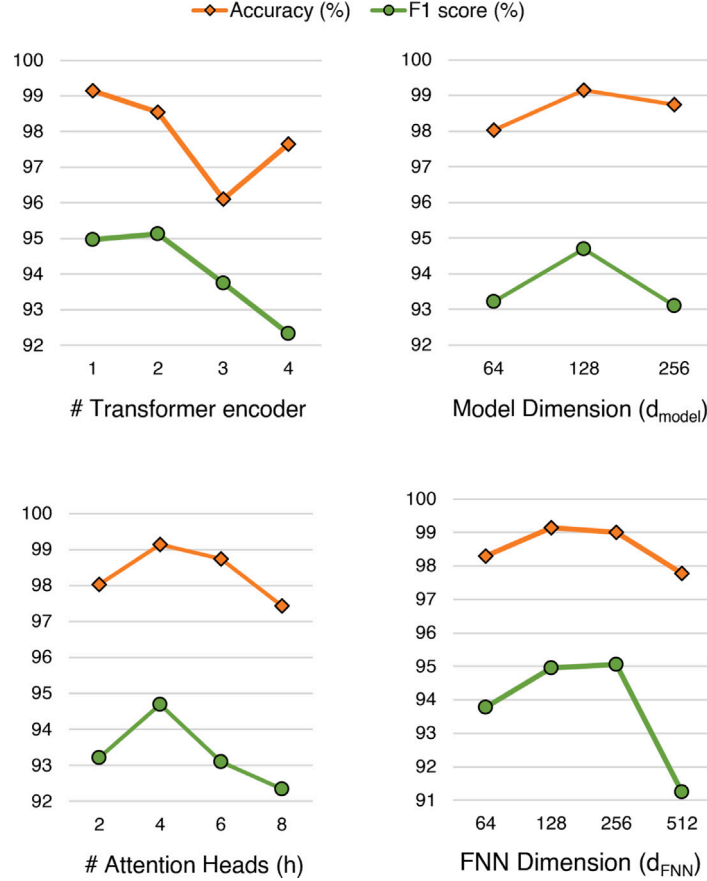
3.4. Computational complexity

The computational complexity (CC) of convolution, recurrent, and self-attention layers is provided in Table 9. While a recurrent layer needs $O(n)$ operations for path length $O(n)$, the attention layer needs

Table 8

Number of total parameters, training time, validation accuracy and loss among some of trained models.

Model details	# Total param.	Training time (s) / Epoch	Max. Val. Acc. (%)	Min. Val. Loss (%)
CNN	381,837	11.5~27.8	99.87	0.76
CNN+LSTM	777,797	21.14~39.16	99.78	0.84
CNN+Att.+LSTM	777,797	30.19~42.23	99.84	0.56
CAT-Net	1,189,637	34.19~47.18	99.89	0.52

**Fig. 11.** Model performance (i.e., accuracy, macro F1 score) variations with transformer encoder hyperparameters.**Table 9**

Computational complexity of different layers.

Layer	Complexity per Layer	Sequential operations	Max path length
Convolution	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Self-attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$

Here, n = input sequences length of transformer encoder, d = model dimension, and k = kernel width of convolution layer. In our model, $n = 128$, $d = 128$, and $k = 21, 23, 25, 27$.

only $O(1)$ operations for path length $O(1)$ due to parallelization. Considering time and space complexity, the self-attention of the transformer encoder presents lower complexity than convolution and recurrent layers, i.e., $CC_{\text{self-attention}} < CC_{\text{convolution}} < CC_{\text{recurrent}}$. Convolution layers make the CAT-Net model computationally more complex than pure transformer-based models, which can only focus on global features. However, the transformer encoder makes the proposed CAT-Net model computationally efficient, with less computational complexity (in both time and space) than pure CNN or CNN-LSTM hybrid models while collecting local and global insight.

3.5. Local and global features

The proposed CAT-Net model efficiently captures local and global ECG characteristics. The first 4 CNN layers focus on ECG signal local insights. Channel attention is employed after each CNN layer to capture the highly relevant parts for arrhythmia classification. The input ECG signal is converted to ECG feature maps in the latent space after passing through four CNN and channel attention layers.

To focus on global dependencies, the ECG feature maps are transferred to the transformer encoder. Following the approach of contextualizing words in NLP, the ECG feature maps are contextualized using the transformer encoder. The multi-head attention layer finds query-key similarities and multiplies the generated attention weights by the value matrix to empower the feature maps for prediction. Overall, the integration of CNN, channel attention, and transformer encoder enables the model to successfully incorporate local as well as global ECG information.

3.6. Effect of small data in 'F' class

The majority of research studies still struggle with the issue of low predictive rate in the 'F' class. In the original MIT-BIH dataset, the F

Table 10
Low predictive rate in 'F' class.

Author	Per-class accuracy (%)					Overall accuracy (%)	Overall macro F1 (%)
	N	S	V	F	Q		
Rafi et al. [48]	99.32	78.46	95.49	71.76	97.79	99.67	98.93
Proposed	99.27	99.60	99.69	99.74	99.98	99.14	94.69
Author	Per-class F1-score (%)					Overall accuracy (%)	Overall macro F1 (%)
	N	S	V	F	Q		
Peng et al. [32]	99.62	91.53	99.26	88.57	99.50	99.28	95.70
Proposed	99.57	92.75	97.80	83.48	99.75	99.14	94.69

Table 11
Comparison of class balancing approaches.

Approach	Required time (Min)	Accuracy (%)	Macro F1 (%)
SMOTE-ENN	42.3	98.60%	94.27%
SMOTE-Tomek	40.8	99.14%	94.69%
ADASYN	37.1	98.02%	93.58%

class contains only 0.7646% data (802 samples out of a total of 104,897 samples). Different class balancing procedures are used to balance the small fraction of data, resulting in a rise in data volume while the information is still the same. Consequently, the prediction in this class became more and more challenging. For example, the model proposed in [48] showed only 71.76% accuracy in the 'F' class and 78.45% accuracy in the 'S' class as illustrated in Table 10. The model [32] showed only 88.57% F1 score in the 'F' class whereas for other classes the score was above 91%. Furthermore, Xia et al. [49] developed a transformer-based model blended with CNN that achieved only 13.04% sensitivity on 'F' class. Similarly, our proposed model CAT-Net predicts this class poorly with an accuracy of 99.74% and F1 score of 83.48%. To enhance the predictive performance in 'F' class, more ECG data of the class is required.

3.7. Effect of class balancing

In order to demonstrate the effect of class imbalance on the outcome of ML algorithms, the proposed model under both balanced and unbalanced training data is analyzed. With imbalanced training data, the model showed only 73.65% F1 score, dropping from 83.48% for the minority class 'F'. Furthermore, the F1 scores of other smaller classes declined by 1% to 6% relative to the balanced model.

The strategy of class balancing also changes the model's performance. The maximum accuracy, macro F1 score and required time for balancing imbalanced ECG training data are listed in Table 11. The model trained by the data balanced by SMOTE-Tomek achieved the highest accuracy and macro F1 score compared to other approaches. Considering the highest performance, SMOTE-Tomek is finally implemented in our CAT-Net model.

3.8. Validation with INCART dataset

In the validation phase of this study, an additional dataset INCART was employed to rigorously assess the robustness and generalization capabilities of the proposed CAT-Net model. Our model achieved 99.58% accuracy and 96.15% macro F1 score for 3-class arrhythmia classification. The confusion matrix and per-class performance score are shown in Table 12. The obtained result is higher than the MIT-BIH data-based model in terms of accuracy and F1 score. This proves the consistent performance of our model. As the local and global contextualized ECG information was captured, the proposed CAT-Net model learns effectively ignoring dataset bias. The robust and dataset-invariant model exhibits promising performance and is applicable to real-world applications.

3.9. Benchmarking with state-of-the-art

To conduct a comprehensive and effective performance evaluation, the proposed model is compared with the existing state-of-the-art methods. Table 13 presents performance metrics for the most representative recent models that employ the same data set, number of classes and almost similar methods of analysis. Recent deep learning based studies focusing specifically on CNN [50], CNN-RNN [51–53], CNN-RNN with attention [30,37], transformer [34,35] and CNN-transformer [36,49,54] were considered. All models used the MIT-BIH dataset except 2 models [35,36], where another dataset [55] was employed. Along with the MIT-BIH dataset, the model [52] used 1-lead Cin-C training 2017 dataset and model [30] used 12 lead CPSC-2018 dataset. The accuracy levels and F1 scores are visualized in ascending order in Figs. 12 and 13, respectively.

The traditional ML models require less data and are computationally cheap. However, the models involve hand-crafted feature extraction, which demands in-depth domain-specific knowledge, resulting in limited performance. Manual feature extraction includes but is not limited to, signal-based (i.e., auto-correlation coefficient power, frequency band power, Shannon's entropy, etc.), statistical (i.e., mean, maximum, variance, skewness, kurtosis, etc.), and morphological features (i.e., R-R interval, QRS duration, etc.). For instance, Venkatesan et al. [56] utilized heart rate variability (HRV) time-frequency-based features with a kNN classifier and achieved 97.5% accuracy for normal vs. abnormal ECG arrhythmia classification. Martis et al. [57] conducted principle component analysis on segmented ECG and obtained 98.11% accuracy for 5 types of arrhythmia classification on the MIT-BIH dataset. Park et al. [58] extracted morphological features such as P wave and QRS complex and used the k-NN classifier and attained an accuracy of 97% in classification.

Nowadays, researchers are focusing on deep learning-based advanced algorithms to ignore the laborious task of manual feature extraction. Gopika et al. proposed the residual convolutional neural network (RCNN) and achieved 4.89% lower accuracy and 0.69% lower F1 score. Xu et al. [52] employed pre-training using the 2017 PhysioNet/CinC Challenge dataset and transfer of parameters during training with the MIT-BIH set. This model considered 4 CLs, 2 residual blocks and 2 BiLSTM layers. The CNN-BiLSTM model yielded the F1 score of 95.92%, but suffered from lower accuracy, 95.9%, which was 3.24% lower than the proposed model.

Wang et al. [50] proposed a model that consists of 33 CNN layers followed by a non-local convolutional block attention module (NCBAM) and that achieved 98.64% accuracy, 96.64% F1 score on the MIT-BIH dataset and 85.07% F1 for the PTB-XL dataset. This model F1 score was 1.95% higher than ours, but its accuracy was lower. Jin et al. [30] proposed the dual-level attentional convolutional long short-term memory (DLA-CLSTM) neural network but its performance was very low 10.38% and 14.15% lower than our method. Essa et al. [53] conducted bagging by ensembling CNN-LSTM and RR intervals and higher-order statistics with LSTM (RRHOS-LSTM) and classified 4 types of arrhythmia. The model accuracy was 95.81% but the highly imbalanced data led to the low F1 score of 71.06%.

Zhao et al. [37] presented an attention-based temporal convolutional network (TCN) and fused it with a mechanism to encode

Table 12
Confusion matrix and per-class performance score of CAT-Net model on INCART testing data (Imbalanced)

	Class type	Predicted class			Per-class score				Total data per-class
		N	S	V	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1 Score (%)	
True class	N	30538	52	34	99.59	99.71	98.74	99.77	30624
	S	32	378	3	99.75	91.53	99.84	89.47	4019
	V	24	2	3993	99.82	99.35	99.88	99.21	413
Overall Performance					99.58	99.58	99.58	99.58 Micro 96.15 Macro 99.58 Weighted	35056

Table 13
Classification performance comparisons on MIT-BIH dataset of state-of-the-art methods.

Authors	Year	# Classes	# Lead(s)	Method(s)	Accuracy (%)	Increment in Accuracy (%)	Macro F1 (%)	Increment in Macro F1 (%)
Yan et al. [34]	2019	4 (N, S, V, F)	1	Transformer + Attention	98.97	0.17	88.73	5.96
Gopika et al. [51]	2020	5	2	RCNN	94.25	4.89	94.00	0.69
Xu et al. [52]	2020	5	1, 2	CNN + BiLSTM	95.9	3.24	95.92	-1.23
Wang et al. [50]	2021	5	12	CNN + Attention	98.64	0.50	96.64	-1.95
Che et al. [36]	2021	9 ^a	12	CNN + Transformer	83.66	15.48	78.60	16.09
Jin et al. [30]	2021	5 (N, A, V, AF, O)	2, 12	DLA-CLSTM	88.76	10.38	80.54	14.15
Essa et al. [53]	2022	4 (N, SVEB, VEB, F)	1	Ensemble of CNN-LSTN & RRHOS-LSTM	95.81	3.33	71.06 ^b	23.63
Meng et al. [35]	2022	3 ^a (N, S, V)	1	Lightweight Transformer	99.32	-0.18	-	-
Yixuan et al. [54]	2022	5	2	STCT: CNN + Transformer	98.96	0.18	99.31^b	-4.62
Zhao et al. [37]	2023	5	1	Attention-based TCN	87.81	11.33	89.46 ^b	5.23
Xia et al. [49]	2023	5	1	(CNN, DAE) + Transformer	97.66	1.48	-	-
Proposed Model (MIT-BIH Dataset)	2023	5	1	CAT-Net: CNN + Attention + Transformer	99.14	4.62% avg.	94.69	6.44% avg.
Proposed Model (INCART Dataset)	2023	3 (N, S, V)	1	CAT-Net: CNN + Attention + Transformer	99.58	-	96.15	-

^a The dataset [55] is used instead of the MIT-BIH dataset.

^b Not specified whether the F1 score is macro or micro.

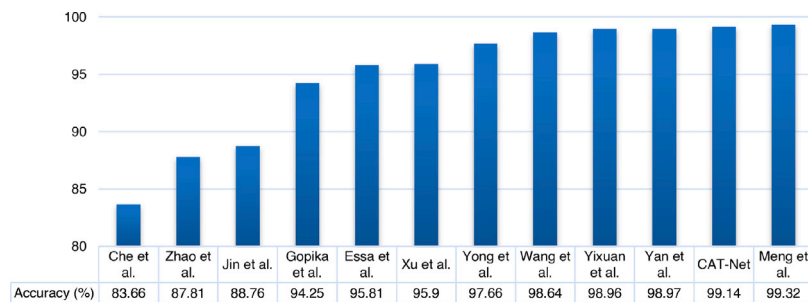


Fig. 12. Comparison in terms of accuracy.

ECG. This model showed good results in an intra-patient scheme with 99.84% accuracy, whereas accuracy degraded to only 87.81% for the inter-patient scenario. This model can predict well specific patient signals but loses performance for other patients. Overall, this model's performance was 11.33% lower in terms of accuracy and 5.23% lower in terms of F1 score.

Yan et al. [34] used a transformer encoder and combined hand-crafted features with transformer-based features for better performance. The model categorized 4 classes of arrhythmias, however, per-class sensitivity and precision were low, causing a low F1 score of 88.73%. To reduce the number of model parameters, Meng et al. [35] proposed a lightweight transformer. The model accuracy of 99.32% was good, however, only 3 types of arrhythmia were classified.

Che et al. [36] combined CNN and transformer and obtained 83.66% accuracy and 78.60% F1 score. They used 7 convolution layers and finally added a transformer encoder along with a link constraint integrated with loss function. The idea of combining CNN and transformer encoder was good, however, model performance was poor. Yixuan et al. [54] proposed Spatial-Temporal Conv-Transformer (STCT) to focus station and temporal information by CNN and transformer encoder respectively. The CNN part is inspired by the architecture of VGGNet. Instead of using FFN, multiple convolution layers were integrated to update the transformer encoder. Xia et al. [49] combined CNN and Denoising Autoencoder (DAE) as well as R-R features and fed these into transformer encoder for 5-class arrhythmia classification. The model achieved 97.66% accuracy.

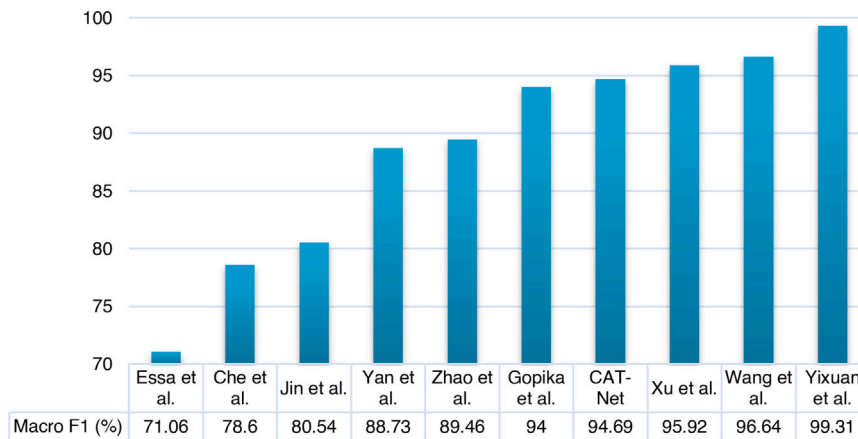


Fig. 13. Comparison in terms of F1 score.

In comparison to the aforementioned models, the proposed CAT-Net model has the highest accuracy of 99.14% for 5-class classification and 99.58% for 3-class classification. Only a few models [50,52,54] exhibit a higher F1 score in comparison to our model. However, these models demonstrate lower accuracy and require the utilization of either 2 or 12 leads of ECG signals. Our model shows a 4.62% average increase in overall accuracy and a 6.44% average increase in the macro F1 score as shown in columns 7 and 9 of Table 13. After all, the impressive accuracy and balanced F1 score for all classes prove the state-of-the-art stature of our model compared to the existing models. An additional important finding of this study is that single-lead ECG data is sufficient for the arrhythmia classification task and this opens the path to adopting machine learning algorithms for mobile and IoT applications that rely only on single-lead ECG devices' recording.

3.10. Limitations and recommendations

A limitation is that the proposed model requires R-peak annotated ECG signals as input. Thus, for real-life applications, an end-to-end machine learning model that includes the R-peak detection step may be necessary. Although this study focused on the challenge of developing a robust and effective model with only single-lead ECG data, additional research challenges of high interest remain in the arrhythmia classification field. First, optimal features from various neural networks equipped with proper attention mechanisms may be fused. Second, IoT and wearable technology-based patient-specific end-to-end models are needed. Third, the major obstacle of lack of annotated and balanced ECG arrhythmia datasets needs to be addressed. Especially, a high volume of 'F' class ECG signals should be recorded to improve the predictive performance of this class.

4. Conclusions

The study proposed a novel CAT-Net model to predict five distinct arrhythmia classes using single-lead ECG signals. By leveraging the capabilities of CNN, channel attention, and transformer encoder architectures, both local and global ECG features were incorporated into the model. It was found that SMOTE-Tomek represents the optimal class balancing approach for ECG data. The model's robustness is validated across two distinct datasets, and it exhibits consistent performance. Using only 1-lead ECG data, the proposed model achieved 99.14% accuracy and 94.69% F1 score, which represent a remarkable performance. This study corroborates that only 1-lead ECG data is adequate to predict arrhythmias provided that the model is properly designed. The proposed model can be used in IoT-based and mobile arrhythmia diagnosis systems due to the low requirement of working in real-time with patients' ECG data captured by only one sensing device. To

facilitate its further implementation and utilization, the comprehensive source code of the CAT-Net model is made accessible through the link https://github.com/rabiul-ai/Arrhythmia_Classification.git.

CRediT authorship contribution statement

Md Rabiul Islam: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Marwa Qaraqe:** Investigation, Validation, Writing – review & editing. **Khalid Qaraqe:** Formal analysis, Investigation. **Erchin Serpedin:** Formal analysis, Investigation, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

We do not have any conflict of interest.

Data availability

Both the MIT-BIH and INCART datasets are publicly available on the website 'PhysioNet'.

Acknowledgment

This work was supported by the Texas A&M University at Qatar (TAMUQ) Research Initiative. Open Access funding provided by the Qatar National Library.

References

- [1] World Health Organization. URL <https://www.who.int/health-topics/cardiovascular-diseases>. (Accessed 8 August 2023).
- [2] E.P.M. Karregat, J.C.L. Himmelreich, W.A.M. Lucassen, W.B. Busschers, H.C.P.M. van Weert, R.E. Harskamp, Evaluation of general practitioners' single-lead electrocardiogram interpretation skills: A case-vignette study, *Fam. Pract.* 38 (2) (2021) 70–75.
- [3] R.G. Afkhami, G. Azarnia, M.A. Tinati, Cardiac arrhythmia classification using statistical and mixture modeling features of ECG signals, *Pattern Recognit. Lett.* 70 (2016) 45–51.
- [4] A.F. Khalaf, M.I. Owis, I.A. Yassine, A novel technique for cardiac arrhythmia classification using spectral correlation and support vector machines, *Expert Syst. Appl.* 42 (21) (2015) 8361–8368.
- [5] M. Kropf, D. Hayn, G. Schreier, ECG classification based on time and frequency domain features using random forests, in: 2017 Computing in Cardiology, *CinC, IEEE*, 2017, pp. 1–4.
- [6] K.N.V.P.S. Rajesh, R. Dhuli, Classification of imbalanced ECG beats using resampling techniques and AdaBoost ensemble classifier, *Biomed. Signal Process. Control* 41 (2018) 242–254.
- [7] H. Shi, H. Wang, Y. Huang, L. Zhao, C. Qin, C. Liu, A hierarchical method based on weighted extreme gradient boosting in ECG heartbeat classification, *Comput. Methods Programs Biomed.* 171 (2019) 1–10.

- [8] N.K. Dewangan, S.P. Shukla, ECG arrhythmia classification using discrete wavelet transform and artificial neural network, in: 2016 IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology, RTEICT, IEEE, 2016, pp. 1892–1896.
- [9] S.-N. Yu, Y.-H. Chen, Electrocardiogram beat classification based on wavelet transformation and probabilistic neural network, *Pattern Recognit. Lett.* 28 (10) (2007) 1142–1150.
- [10] Z. Zhang, J. Dong, X. Luo, K.-S. Choi, X. Wu, Heartbeat classification using disease-specific feature selection, *Comput. Biol. Med.* 46 (2014) 79–89.
- [11] E.B. Mazomenos, D. Biswas, A. Acharyya, T. Chen, K. Maharatna, J. Rosengarten, J. Morgan, N. Curzen, A low-complexity ECG feature extraction algorithm for mobile healthcare applications, *IEEE J. Biomed. Health Inf.* 17 (2) (2013) 459–469.
- [12] M. Lagerholm, C. Peterson, G. Braccini, L. Edenbrandt, L. Sornmo, Clustering ECG complexes using Hermite functions and self-organizing maps, *IEEE Trans. Biomed. Eng.* 47 (7) (2000) 838–848.
- [13] P. Melin, J. Amezcua, F. Valdez, O. Castillo, A new neural network model based on the LVQ algorithm for multi-class classification of arrhythmias, *Inform. Sci.* 279 (2014) 483–497.
- [14] I. Christov, G. Gómez-Herrero, V. Krasteva, I. Jekova, A. Gotchev, K. Egiazarian, Comparative study of morphological and time-frequency ECG descriptors for heartbeat classification, *Med. Eng. Phys.* 28 (9) (2006) 876–887.
- [15] U.R. Acharya, H. Fujita, M. Adam, O.S. Lih, T.J. Hong, V.K. Sudarshan, J.E.W. Koh, Automated characterization of arrhythmias using nonlinear features from tachycardia ECG beats, in: 2016 IEEE International Conference on Systems, Man, and Cybernetics, SMC, IEEE, 2016, pp. 000533–000538.
- [16] S. Kiranyaz, T. Ince, M. Gabbouj, Real-time patient-specific ECG classification by 1-D convolutional neural networks, *IEEE Trans. Biomed. Eng.* 63 (3) (2015) 664–675.
- [17] U.R. Acharya, et al., Application of higher-order spectra for the characterization of coronary artery disease using electrocardiogram signals, *Biomed. Signal Process. Control* 31 (2017) 31–43.
- [18] P.M. Tripathi, A. Kumar, M. Kumar, R. Komaragiri, Multilevel classification and detection of cardiac arrhythmias with high-resolution superlet transform and deep convolution neural network, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–13.
- [19] M. Hammad, A.M. Ilyasu, A. Subasi, E.S.L. Ho, A.A. Abd El-Latif, A multitier deep learning model for arrhythmia detection, *IEEE Trans. Instrum. Meas.* 70 (2020) 1–9.
- [20] O. Faust, Y. Hagiwara, T.J. Hong, O.S. Lih, U.R. Acharya, Deep learning for healthcare applications based on physiological signals: A review, *Comput. Methods Programs Biomed.* 161 (2018) 1–13.
- [21] P. Rajpurkar, A.Y. Hannun, M. Haghighpanahi, C. Bourn, A.Y. Ng, Cardiologist-level arrhythmia detection with convolutional neural networks, 2017, arXiv preprint arXiv:1707.01836.
- [22] J.H. Tan, Y. Hagiwara, W. Pang, I. Lim, S.L. Oh, M. Adam, R. San Tan, M. Chen, U.R. Acharya, Application of stacked convolutional and long short-term memory network for accurate identification of CAD ECG signals, *Comput. Biol. Med.* 94 (2018) 19–26.
- [23] S.L. Oh, E.Y.K. Ng, R. San Tan, U.R. Acharya, Automated diagnosis of arrhythmia using combination of CNN and LSTM techniques with variable length heart beats, *Comput. Biol. Med.* 102 (2018) 278–287.
- [24] Q. Yao, X. Fan, Y. Cai, R. Wang, L. Yin, Y. Li, Time-incremental convolutional neural network for arrhythmia detection in varied-length electrocardiogram, in: IEEE 16th Intl. Conf. on Dependable, Autonomic and Secure Computing, 16th Intl. Conf. on Pervasive Intelligence and Computing, 4th Intl. Conf. on Big Data Intelligence and Computing and Cyber Science and Technology Congress, DASC/PiCom/DataCom/CyberSciTech, IEEE, 2018, pp. 754–761.
- [25] E. Prabhakararao, S. Dandapat, Myocardial infarction severity stages classification from ECG signals using attentional recurrent neural network, *IEEE Sens. J.* 20 (15) (2020) 8711–8720.
- [26] P. Singh, A. Sharma, Attention-based convolutional denoising autoencoder for two-lead ECG denoising and arrhythmia classification, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–10, Art. no. 4007710.
- [27] Q. Yao, R. Wang, X. Fan, J. Liu, Y. Li, Multi-class arrhythmia detection from 12-lead varied-length ECG using attention-based time-incremental convolutional neural network, *Inf. Fusion* 53 (2020) 174–182.
- [28] J. Zhang, A. Liu, M. Gao, X. Chen, X. Zhang, X. Chen, ECG-based multi-class arrhythmia detection using spatio-temporal attention-based convolutional recurrent neural network, *Artif. Intell. Med.* 106 (2020) 101856.
- [29] W. Ullah, I. Siddique, R.M. Zulqarnain, M.M. Alam, I. Ahmad, U.A. Raza, Classification of arrhythmia in heartbeat detection using deep learning, *Comput. Intell. Neurosci.* 2021 (2021) 1–13, Art. ID 2195922.
- [30] Y. Jin, J. Liu, Y. Liu, C. Qin, Z. Li, D. Xiao, L. Zhao, C. Liu, A novel interpretable method based on dual-level attentional deep neural network for actual multilabel arrhythmia detection, *IEEE Trans. Instrum. Meas.* 71 (2021) 1–11, Art. no. 2500311.
- [31] E. Prabhakararao, S. Dandapat, Multi-scale convolutional neural network ensemble for multi-class arrhythmia classification, *IEEE J. Biomed. Health Inf.* 26 (8) (2021) 3802–3812.
- [32] X. Peng, W. Shu, C. Pan, Z. Ke, H. Zhu, X. Zhou, W.W. Song, DSCSSA: A classification framework for spatiotemporal features extraction of arrhythmia based on the Seq2Seq model with attention mechanism, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–12, Art. no. 2515112.
- [33] R. Hu, J. Chen, L. Zhou, A transformer-based deep neural network for arrhythmia detection using continuous ECG signals, *Comput. Biol. Med.* 144 (May 2022) 105325.
- [34] G. Yan, S. Liang, Y. Zhang, F. Liu, Fusing transformer model with temporal features for ECG heartbeat classification, in: 2019 IEEE International Conference on Bioinformatics and Biomedicine, BIBM, IEEE, 2019, pp. 898–905.
- [35] L. Meng, W. Tan, J. Ma, R. Wang, X. Yin, Y. Zhang, Enhancing dynamic ECG heartbeat classification with lightweight transformer model, *Artif. Intell. Med.* 124 (2022) 102236.
- [36] C. Che, P. Zhang, M. Zhu, Y. Qu, B. Jin, Constrained transformer network for ECG signal processing and arrhythmia classification, *BMC Med. Inform. Decis. Mak.* 21 (184) (2021) 1–13.
- [37] Y. Zhao, J. Ren, B. Zhang, J. Wu, Y. Lyu, An explainable attention-based TCN heartbeats classification model for arrhythmia detection, *Biomed. Signal Process. Control* 80, Part 1 (2023) 104337.
- [38] S.M. Mathews, C. Kambhampettu, K.E. Barner, A novel application of deep learning for single-lead ECG classification, *Comput. Biol. Med.* 99 (2018) 53–62.
- [39] ANSI/AAMI, Testing and reporting performance results of cardiac rhythm and ST segment measurement algorithms, Assoc. Adv. Med. Instrum. (1998) document ANSI/AAMI/ISO EC57, 1998-(R), American National Standard Institute, Inc. (ANSI), 2008.
- [40] G.B. Moody, R.G. Mark, The impact of the MIT-BIH arrhythmia database, *IEEE Eng. Med. Biol. Mag.* 20 (3) (2001) 45–50.
- [41] St Petersburg INCART 12-lead Arrhythmia Database. URL <https://physionet.org/content/incartdb/1.0.0/>. (Accessed 3 July 2023).
- [42] L. Sathiyapriya, L. Murali, T. Manigandan, Analysis and detection R-peak detection using modified Pan-Tompkins algorithm, in: 2014 IEEE International Conference on Advanced Communications, Control and Computing Technologies, IEEE, 2014, pp. 483–487.
- [43] G.E. Batista, R.C. Prati, M.C. Monard, A study of the behavior of several methods for balancing machine learning training data, *ACM SIGKDD Explor. Newsl.* 6 (1) (2004) 20–29.
- [44] N.V. Chawla, K.W. Bowyer, L.O. Hall, W.P. Kegelmeyer, SMOTE: Synthetic minority over-sampling technique, *J. Artificial Intelligence Res.* 16 (2002) 321–357.
- [45] H. He, Y. Bai, E.A. Garcia, S. Li, ADASYN: adaptive synthetic sampling approach for imbalanced learning, in: 2008 IEEE International Joint Conference on Neural Networks, IEEE World Congress on Computational Intelligence, IEEE, 2008, pp. 1322–1328.
- [46] S. Woo, J. Park, J.-Y. Lee, I.S. Kweon, CBAM: Convolutional block attention module, in: Proceedings of the European Conference on Computer Vision, ECCV, 2018, pp. 3–19.
- [47] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [48] T.H. Rafi, Y.W. Ko, HeartNet: Self multihead attention mechanism via convolutional network with adversarial data synthesis for ECG-based arrhythmia classification, *IEEE Access* 10 (2022) 100501–100512.
- [49] Y. Xia, Y. Xiong, K. Wang, A transformer model blended with CNN and denoising autoencoder for inter-patient ECG arrhythmia classification, *Biomed. Signal Process. Control* 86 (2023) 105271.
- [50] J. Wang, X. Qiao, C. Liu, X. Wang, Y. Liu, L. Yao, H. Zhang, Automated ECG classification using a non-local convolutional block attention module, *Comput. Methods Programs Biomed.* 203 (May 2021) 106006.
- [51] P. Gopika, V. Sowmya, E. Gopalakrishnan, K. Soman, Transferable approach for cardiac disease classification using deep learning, in: Deep Learning Techniques for Biomedical and Health Informatics, Elsevier, 2020, pp. 285–303.
- [52] X. Xu, S. Jeong, J. Li, Interpretation of electrocardiogram (ECG) rhythm by combined CNN and BiLSTM, *IEEE Access* 8 (2020) 125380–125388.
- [53] E. Essa, X. Xie, An ensemble of deep learning-based multi-model for ECG heartbeats arrhythmia classification, *IEEE Access* 9 (2021) 103452–103464.
- [54] Y. Qiu, W. Chen, L. Yue, M. Xu, B. Zhu, STCT: Spatial-temporal conv-transformer network for cardiac arrhythmias recognition, in: International Conference on Advanced Data Mining and Applications, Springer, 2022, pp. 86–100.
- [55] Z. Cai, C. Liu, H. Gao, X. Wang, L. Zhao, Q. Shen, E.Y.K. Ng, J. Li, An open-access long-term wearable ECG database for premature ventricular contractions and supraventricular premature beat detection, *J. Med. Imag. Health Inform.* 10 (11) (2020) 2663–2667.
- [56] C. Venkatesan, P. Karthigaikumar, R. Varatharajan, A novel LMS algorithm for ECG signal preprocessing and KNN classifier based abnormality detection, *Multimedia Tools Appl.* 77 (2018) 10365–10374.
- [57] R.J. Martis, U.R. Acharya, K.M. Mandana, A.K. Ray, C. Chakraborty, Application of principal component analysis to ECG signals for automated diagnosis of cardiac health, *Expert Syst. Appl.* 39 (14) (2012) 11792–11800.
- [58] J. Park, K. Lee, K. Kang, Arrhythmia detection from heartbeat using k-nearest neighbor classifier, in: 2013 IEEE International Conference on Bioinformatics and Biomedicine, IEEE, 2013, pp. 15–22.